Classification Rules

Manchester-based Clinical-Functional Classification of Emergency Department Presenting Complaints

(EDEN-RICAPPS project)

Purpose of this document

This document describes the **conceptual and clinical rules** underlying the Manchester-based clinical-functional classification of emergency department presenting complaints provided in the dataset `classification_manchester_ed_presenting_complaints.xlsx`.

The objective of this classification is to support **early risk stratification** in emergency care using information available at initial presentation, rather than to represent definitive diagnoses or outcomes.

Conceptual framework

The classification is inspired by principles of the **Manchester Triage System** but does **not** aim to reproduce or replace any official Manchester algorithms. Instead, it provides a **simplified clinical-functional grouping** suitable for observational research and methodological analyses.

Key principles:

- Classification is based on the **presenting complaint or diagnostic code** recorded at emergency department arrival.

- Each episode is assigned to **one and only one** clinical-functional chapter.
- The classification reflects **initial functional risk**, not final diagnosis.
- Categories are defined **a priori**, independent of outcomes.

Clinical-functional chapters

The system defines eight mutually exclusive chapters:

1. Cardiorespiratory
Presenting complaints related to the cardiovascular or respiratory systems, including chest pain, dyspnea, palpitations, syncope, and other symptoms suggesting potential cardiorespiratory compromise.

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2. Acute neurological
Presenting complaints indicating acute neurological dysfunction, such as altered level of consciousness, focal neurological deficits, seizures, acute headache, or suspected central nervous system pathology.

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3. Sepsis / potential infection
Complaints suggestive of systemic infection or inflammatory response, including fever, suspected sepsis, severe infections, and related systemic presentations.

This category also includes conditions with **metabolic or hematological impact** when the functional presentation is systemic rather than localized.

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4. Medical–surgical abdominal

Presenting complaints related to abdominal pain or gastrointestinal symptoms with potential medical or surgical relevance, including gastrointestinal bleeding and other acute abdominal syndromes.

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5. Major trauma

Presentations related to high-energy trauma or mechanisms with potential for severe injury, such as traffic accidents, significant falls, or polytrauma.

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6. Debility and fatigue

A conceptual category intended to capture **general functional deterioration** or weakness that does not clearly fit within other acute syndromic categories.

This chapter is intentionally defined even if it may be **empty in some cohorts**, as many cases of debility are commonly coded under more specific etiological diagnoses (e.g., metabolic, hematological, or neurological causes).

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7. Initial low vital risk

Presenting complaints with low immediate risk to vital functions at initial assessment, including musculoskeletal pain, minor symptoms, psychiatric presentations without acute medical compromise, and similar presentations.

This category often functions as a **reference group** in analyses.

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8. Other complaints

A residual category for presenting complaints that cannot be reasonably assigned to any of the above clinical-functional chapters.

This category is intended to ensure **completeness of classification** and should be used sparingly.

Assignment rules

- Each diagnostic or presenting-complaint code must be assigned to **one single chapter**.
- Assignment is based on the **primary functional presentation**, not on comorbidities or secondary findings.
- When multiple interpretations are possible, the chapter reflecting the **highest initial functional risk** is preferred.

Limitations

- The classification does not capture **clinical severity gradients** within categories.
- It is not a triage algorithm and should not be used for bedside clinical decision-making.
- Differences in local coding practices may affect category distributions across datasets.

Scope and reuse

This classification is intended for:

- observational research in emergency medicine,
- early risk stratification studies,
- methodological and epidemiological analyses,
- quality and safety research.

It is not intended to replace formal triage systems or clinical judgment.

Related resources

- **Project:** EDEN-RICAPPS
- **Protocol registration:** ClinicalTrials.gov (NCT07194122)

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Citation

Please cite the Zenodo record associated with this dataset when using or adapting these classification rules in derived works.